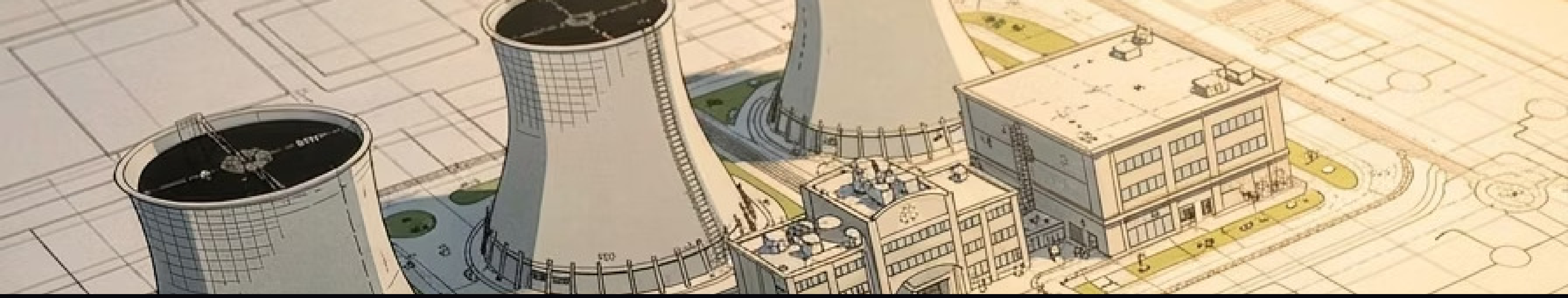
Half-Life: A Comparative Study of Two Nuclear Disasters

Two of the most consequential nuclear disasters in history, their causes, environmental impacts, key personnel, liabilities, and the screen works they inspired.



FUKUSHIMA DAIICHI

Fukushima Daiichi Nuclear Power Plant

Fukushima Daiichi was one of the largest nuclear power stations in the world before the 2011 disaster. Understanding its history, design, and operational context is essential to understanding how the accident unfolded and why its consequences were so severe.

Location and Geography

Fukushima Daiichi is located on the Pacific coast of Fukushima Prefecture, in the Okuma and Futaba towns of the Hamadori region of northeastern Honshu, Japan. The site sits approximately 220 km northeast of Tokyo and around 10 km north of the town of Tomioka. It was deliberately sited on the coast to allow seawater intake for reactor cooling. The plant sits on a bluff that was originally 35 metres above sea level; during construction, the site was excavated down to approximately 10 metres above sea level to reduce the height that seawater pumps needed to lift cooling water. This decision later proved critical, as it reduced the effective height of the seawall relative to the tsunami that struck in 2011.

Establishment and Construction

The plant was developed by Tokyo Electric Power Company (TEPCO) and construction began in 1967. Unit 1, a boiling water reactor (BWR) designed by General Electric, began commercial operation in March 1971, making it one of Japan's earliest commercial nuclear reactors. Units 2 through 6 were added progressively between 1974 and 1979. At its peak, the six-unit plant had a combined generating capacity of approximately 4,696 megawatts, supplying electricity to the greater Tokyo metropolitan area. A sister plant, Fukushima Daini, was constructed nearby and also operated by TEPCO.

Reactor Design: Boiling Water Reactors (BWR)

All six units at Fukushima Daiichi used boiling water reactor technology, in which water is boiled directly in the reactor pressure vessel to produce steam that drives turbines. Units 1 through 5 used Mark I containment structures, a design developed by General Electric in the 1960s. The Mark I containment was later identified as having limited capacity to manage hydrogen buildup and overpressure events, a vulnerability that contributed directly to the 2011 explosions. Unit 6 used a Mark II containment design.

Operator: Tokyo Electric Power Company (TEPCO)

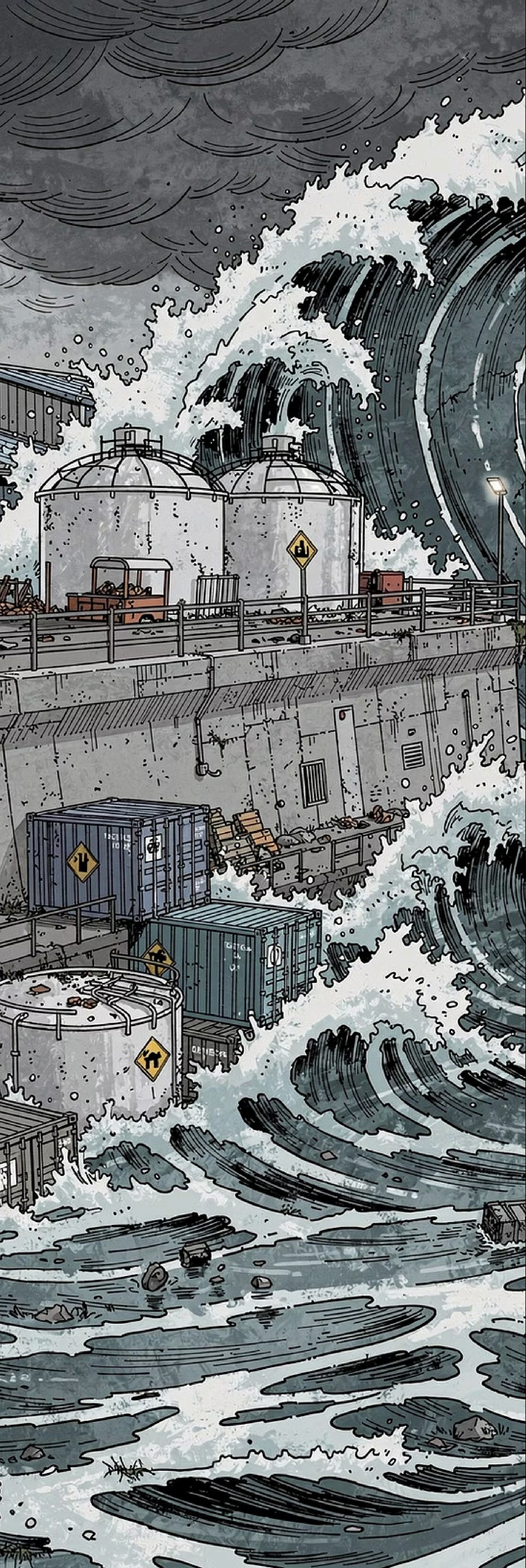
TEPCO was Japan's largest private electric utility and one of the largest in the world, serving approximately 29 million customers across the Kanto region. It held operating licences for Fukushima Daiichi and was responsible for all aspects of plant safety, maintenance, staffing, and regulatory compliance. TEPCO's relationship with Japan's nuclear regulatory framework was later described by investigators as insufficiently adversarial, with regulators and the operator sharing assumptions about risk that were not adequately tested.

Workforce and Operations

The plant employed several thousand workers directly and through contractor firms. Day-to-day operations were managed by a plant superintendent, with TEPCO's headquarters in Tokyo maintaining oversight. At the time of the 2011 accident, the plant superintendent was Masao Yoshida, who had taken up the role in June 2010. Routine maintenance, refuelling, and safety inspections were conducted on a rolling schedule across the six units, with some units offline for scheduled work at the time of the earthquake.

Pre-Accident Safety Record and Warnings

Fukushima Daiichi had a mixed safety record. In 2002, TEPCO admitted to falsifying safety inspection records across multiple plants, including Fukushima Daiichi, leading to temporary shutdowns and management changes. In the years before 2011, Japanese and international researchers had raised concerns about the adequacy of the seawall against large tsunamis. Internal TEPCO documents later revealed that some engineers had flagged tsunami risk as early as 2008, but no significant upgrades were made before the disaster.



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Overview: The 2011 Disaster

On 11 March 2011, the Great East Japan Earthquake and subsequent tsunami caused a severe nuclear accident at the Fukushima Daiichi nuclear power plant, resulting in core damage in three reactors and significant releases of radioactive material to the atmosphere and the Pacific Ocean. The fallout led to mass evacuations, long-term land contamination, exclusion zones, decades-long decommissioning work, and a large compensation and liability regime centered on Tokyo Electric Power Company (TEPCO).

Eight Key Events Leading to the Fallout

1. **Earthquake**

   Magnitude 9.0 quake strikes off Honshu, shutting down reactors and cutting off-site power.

2. **Tsunami**

   Waves 14 to 15 m high overtop the seawall, flooding turbine buildings and electrical equipment.

3. **Station Blackout**

   Flooding disables emergency diesel generators, leaving the site without power for cooling.

4. **Core Damage**

   Water levels drop in Units 1 to 3; cores overheat and suffer meltdowns, generating hydrogen.

5. **Explosions**

   Hydrogen explosions in Units 1, 3, and 4 destroy building superstructures and obstruct response.

6. **Evacuation**

   Government expands evacuation orders and establishes a 20 km no-entry zone.